

Storm Sequence Generator

Multi-Storm Sequence Model v3.0 (MKM-SS-001)

MKM Research Labs

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1 Executive Summary

The Storm Sequence Generator draws the synthetic storm catalogue for the multi-storm stress test: by default 10,000 `StormSequence`s, each holding 1–5 storms within a 168-hour event window. All rain ends by hour 156, which leaves at least 12 hours to watch the rivers drain. The window follows the hours clause in catastrophe reinsurance. The stress pipeline turns each sequence into a hydrograph at every gauge. Those hydrographs train the per-gauge flood classifiers (MKM-FC-001) and drive the Portfolio Stress scenarios.

There are four sequence types:

- **isolated:** one storm;
- **doublet:** two storms with a medium gap of 24–72 h;
- **cluster:** three or four storms with medium gaps;
- **persistent:** four or five storms with short gaps of 6–36 h.

The chance of a multi-storm sequence rises with the intensity category, from 5% for minimal storms to 85% for catastrophic ones.

v3.0 keeps a gap between every pair of storms. v2.0 capped each storm at $156/n$ h before drawing the gaps. Long storms then filled the window and left no time between them: in 46% of multi-storm sequences the storms ran back to back, so the catchment recovery the gaps exist to model did not happen. v3.0 reserves each gap's minimum (6 h short, 24 h medium) before it caps the storms, and shortens a long gap only toward that minimum. Every multi-storm sequence now has a gap of at least its minimum, and every sequence still fits the window without retries.

v3.0 also corrects what v2.0 said. The Wolfram benchmark of v2.0 (v0.1) found that the code computed v2.0's equations but that 11 of its 17 claims did not hold:

- its cap and scaling rates and its Table 9 were wrong;
- its intensity correlation was misdescribed;
- its base precipitations were for catchments the platform does not have;
- it said no runtime override existed, but one does, and it reached the generator only in part;
- its paths, command line and classifier evidence were stale.

v3.0 moves every table to `config.port`, where an override reaches the generator. Every figure here is generated from the code. Wolfram benchmark v0.2 tests each claim (validation question Q8).

Key risk: the durations do not fit the window. Table 3's storms are too long for three to five of them, with their gaps, to fit in 156 h. The cap binds on 60.5% of all duration draws (§6.2), and in 63.4% of multi-storm sequences every storm is capped. This exceeds the 50% level at which this document asks for recalibration (Q6). Recalibrating Tables 1 and 3 is outstanding (Q9). The classifiers on file were trained on v2.0's sequences and must be retrained (§7.3).

2 Model Purpose and Scope

2.1 Purpose

1. **Synthetic sequences.** A catalogue of storm sequences across the moderate to catastrophic categories, each with 1–5 timed storms.
2. **Hydrograph timing.** Each storm's start, duration, intensity and peak position, so the stress

pipeline can compose multi-peak hydrographs that show a catchment recovering between storms.

3. **Classifier training data.** Hydrographs with antecedent wetness: a gauge that has only partly recovered from one storm is at greater risk from the next.

2.2 Where it is used

- `app/commands/port/stages/storm.py`, the `--stressm` stage of `phys.py port`, calls the stress pipeline.
- `src/port/src/stressm/pipeline/orchestrator/_core.py` generates the sequences, saves them, and computes each gauge's response.
- `src/port/src/stressm/batch_train.py` and `src/routes/trading/stress/training.py` load the saved sequences to train the classifiers (MKM-FC-001).
- The Portfolio Stress screens read the sequences and the classifiers' outputs.

2.3 Scope

In scope: sequence type, storm count, durations, gaps, intensities, precipitation, the window, and batch generation. Out of scope: the spatial footprint (MKM-SG-001), the gauge response and hydrology in the package's `models/` subpackage, hazard curves (MKM-GH-001), pricing (MKM-PR-001), climate projections and tides.

2.4 Tier

Tier 2. The generator produces no regulated output itself, but it is the main input to the stress classifiers, which feed the Portfolio Stress P&L.

3 Mathematical Framework

3.1 Event Window

The event window W and the hour W_P by which all precipitation must end are

$$W = 168 \text{ h}, \tag{1}$$

$$W_P = W - W_{\text{drain}} = 168 - 12 = 156 \text{ h}. \tag{2}$$

A sequence's own drainage window runs from the end of its last storm to the end of the event window:

$$W_{\text{drain}}^{(\text{seq})} = W - (t_{\text{start},n} + d_n) \;\geq\; W_{\text{drain}} = 12 \text{ h}. \tag{3}$$

3.2 Sequence Types

The storm count for a cluster or persistent sequence is drawn uniformly from its range.

Table 1: Sequence types.

Type	Storms	Gap type	Physical analogue
isolated	1	—	a single frontal system
doublet	2	medium	an Atlantic low followed by a secondary low
cluster	3–4	medium	an active frontal passage
persistent	4–5	short	a blocking pattern; near-continuous rain

3.3 Sequence Type Selection

A storm becomes a multi-storm sequence with a probability that depends on its category:

$$P(\text{sequence}) = \begin{cases} 0.05 & \text{minimal} \\ 0.15 & \text{baseline} \\ 0.30 & \text{moderate} \\ 0.50 & \text{severe} \\ 0.70 & \text{extreme} \\ 0.85 & \text{catastrophic} \end{cases} \tag{4}$$

If it does, the type is drawn from Table 2.

Table 2: Type weights for a multi-storm sequence, by category.

Category	Doublet	Cluster	Persistent
minimal, baseline	0.60	0.30	0.10
moderate	0.50	0.35	0.15
severe	0.30	0.50	0.20
extreme	0.20	0.45	0.35
catastrophic	0.15	0.45	0.40

3.4 Storm Durations

Each storm's duration is a triangular draw on its category's parameters (Table 3), rounded to the nearest whole hour:

$$D = \text{round}\left(\text{Triangular}(a_{\text{cat}}, c_{\text{cat}}, b_{\text{cat}})\right). \tag{5}$$

Table 3: Storm duration (hours), triangular, by category.

Category	Min	Mode	Max
minimal	2	6	12
baseline	6	12	24
moderate	12	30	60
severe	24	48	96
extreme	36	72	144
catastrophic	48	120	240

3.4.1 Duration Cap

Each of the $n-1$ gaps is guaranteed at least its gap type's minimum $g_{\min}$ (6 h short, 24 h medium; Table 4). The window reserves that time first, and divides what is left equally between the storms:

$$d_i = \min\left(D_i,\ \frac{W_P - (n-1)\,g_{\min}}{n}\right), \qquad i = 1, \ldots, n. \tag{6}$$

The cap is 156 h for an isolated storm, 66 h in a doublet, 36 h and 21 h in clusters of three and four, and 34.5 h and 26.4 h in persistent sequences of four and five. If the window cannot hold n storms and their minimum gaps (cap ≤ 0, possible only under an override), generation stops with an error. A capped duration is not rounded.

3.5 Inter-Storm Gaps

Gaps are triangular draws on their gap type's parameters, rounded to the nearest whole hour.

Table 4: Inter-storm gap (hours), triangular, by gap type.

Gap type	Used by	Min	Mode	Max	Interpretation
short	persistent	6	18	36	minimal drainage; high compounding
medium	doublet, cluster	24	48	72	partial drainage (30–60% recovery)
long	—	48	96	144	substantial drainage; unused

3.5.1 Gap Shortening

Let $R = W_P - \sum_i d_i$ be the time the storms leave. Eq. (6) guarantees $R \geq (n-1)\,g_{\min}$. If the raw gaps $G_1, \ldots, G_{n-1}$ overflow it,

$$\sum_{j=1}^{n-1} G_j > R, \tag{7}$$

each gap keeps its minimum and the time above the minimum is scaled to fit:

$$g_j = g_{\min} + (G_j - g_{\min})\,\frac{R - (n-1)\,g_{\min}}{\sum_{k=1}^{n-1}(G_k - g_{\min})}, \qquad j = 1, \ldots, n-1. \tag{8}$$

Otherwise the raw gaps are used. Either way $g_j \geq g_{\min}$ and $\sum d_i + \sum g_j \leq W_P$. When every storm is capped, $R = (n-1)\,g_{\min}$ and every gap is held at its minimum.

3.6 Storm Placement

$$t_{\text{start},1} = 0, \tag{9}$$

$$t_{\text{start},i} = t_{\text{start},i-1} + d_{i-1} + g_{i-1}, \qquad i = 2, \ldots, n. \tag{10}$$

3.7 Intensities

The sequence's base intensity is drawn per category (Table 5) and floored:

$$F_0 = \max(0.1,\ \mathcal{N}(\mu_{\text{cat}}, \sigma^2_{\text{cat}})). \tag{11}$$

Table 5: Base intensity, by category.

Category	μ	σ
minimal	0.30	0.10
baseline	0.60	0.15
moderate	1.00	0.20
severe	1.80	0.30
extreme	3.00	0.50
catastrophic	5.00	0.80

Each storm varies around F_0 by δ, within $\pm v = \pm 0.20$. The first storm varies downward:

$$F^{(1)} = \max\big(0.1,\ F_0(1+\delta_1)\big), \qquad \delta_1 \sim \text{Uniform}(-0.2, 0). \tag{12}$$

The code draws δ_1 and then, with probability 0.70, draws it again from the same law. The redraw cannot change δ_1's law. Its configuration parameter, `FIRST_STORM_DOMINANT_PROB` (0.30, shown on the Trading Control screen), therefore has no effect on the output distribution. It is kept only so the random stream is unchanged. Each later storm varies up with probability $p_{\text{corr}} = 0.70$:

$$F^{(i)} = \max\big(0.1,\ F_0(1+\delta_i)\big), \qquad \delta_i \sim \begin{cases} \text{Uniform}(0, 0.2) & \text{with probability } 0.70, \\ \text{Uniform}(-0.2, 0) & \text{with probability } 0.30. \end{cases} \tag{13}$$

A later storm is then at least as strong as the first with probability $p_{\text{corr}} + (1 - p_{\text{corr}})/2 = 0.85$ (above the floor): it is if it varies up, and half the time if both vary down. The model represents antecedent wetness this way: a catchment wet from one storm responds more to the next.

3.8 Precipitation

$$P_i = P_{\text{base}}\, F^{(i)}, \tag{14}$$

$$\text{peak position} \sim \text{Beta}(2, 2), \tag{15}$$

where P_{base} is the catchment's base precipitation (Table 7) and the peak position is the fraction of the storm's duration at which it peaks.

3.9 Aggregates

$$P_{\text{total}} = \textstyle\sum_i P_i, \tag{16}$$

$$F_{\max} = \max_i F^{(i)}, \tag{17}$$

$$\bar{F} = \tfrac{1}{n} \textstyle\sum_i F^{(i)}, \tag{18}$$

$$F_{\text{cum}} = \textstyle\sum_i F^{(i)}, \tag{19}$$

$$D_{\text{total}} = \textstyle\sum_i d_i + \sum_j g_j. \tag{20}$$

F_{cum} carries the compounding of several storms into the stress test, where the peak alone would not.

4 Input Parameters and Calibration

4.1 Batch Generation

`generate_event_set(count, catchment_id, intensity_weights, force_sequence_type, seed)` draws each sequence's category from Table 6 (normalised, so custom weights need not sum to one). The default count is 10,000.

Table 6: Default category weights.

Category	Weight
moderate	0.40
severe	0.35
extreme	0.20
catastrophic	0.05

4.2 Catchment Base Precipitation

P_{base} is `BASE_PRECIPITATION_MM` in the catchment's storm module, `catch/<catchment>/storm.py`. Three catchments have one (Table 7). A catchment without one (the single-module catchments under `catch/`, such as the Rhine) takes the Thames's 35 mm.

Table 7: Base precipitation by catchment.

Catchment	P_{base} (mm)
thames	35.0
mekong	90.0
halong	100.0
any other	35.0

4.3 Constants

Table 8: Constants.

Constant	Symbol	Value	Meaning
Event window	W	168 h	the hours clause
Minimum drainage	W_{drain}	12 h	watched after the last storm
Precipitation window	W_P	156 h	$W - W_{\text{drain}}$
Intensity variation	v	0.20	each storm's range around F_0
Correlation probability	p_{corr}	0.70	$P(\delta_i \geq 0)$ for a later storm
Later storm $\geq$ first		0.85	$p_{\text{corr}} + (1 - p_{\text{corr}})/2$
Peak shape	(α, β)	(2, 2)	peak position
Intensity floor		0.1	on F_0 and every $F^{(i)}$

4.4 Where the Parameters Live, and Overrides

Every parameter in Tables 2–8, except P_{base}, is in `config/port/_storm.py` (`config.port`). The generators read them from there on every draw. `storm_control.json`, in the catchment's input directory, overrides any of them:

- The Trading Control screen writes the file and applies it when it saves.

- Importing `config` applies it too, so a command-line run uses it.

Because the generators read `config.port` on every draw, an override takes effect whether it was applied before or after the generator was imported. A file that sets some rows of a table changes those rows and keeps the others. Saving on the screen requires the trading write permission. `force_sequence_type` forces every sequence's type, for testing.

5 Implementation Details

5.1 Software Architecture

The package is `src/port/src/storm_multi/`:

- `core/data_structures.py`: `SequenceStorm`, `StormSequence`, `SequenceType`, `GapType`, `make_storm_id`, `make_sequence_id`.
- `generators/sequence_generator.py`: `SequenceGenerator`, which applies Eqs. (6)–(20).
- `generators/batch_generator.py`: `generate_event_set`.
- `generators/duration_sampler.py`: `sample_duration`, `get_duration_range`.
- `generators/gap_sampler.py`: `sample_gap`, `get_gap_range`, `get_min_gap`.
- `generators/intensity_sampler.py`: `should_generate_sequence`, `sample_sequence_type`, `get_storm_count`, `sample_storm_intensity`.
- `utils/validation.py`: `validate_sequence`, `validate_event_set`, `fits_in_window`, `precip_end_hour`, `min_gap_hours`.
- `utils/serialization.py`: `save_sequences`, `save_summary`, `load_sequences`.

The `models/` subpackage (sequence response, hydrology, spatial correlation) is part of the gauge response, not of this model.

5.2 Generation Flow

For each of the N sequences, `generate_event_set`:

1. draws the category (Table 6) and F_0 (Eq. (11));
2. draws the type (Eq. (4), Table 2) and the storm count;
3. draws each storm's duration and caps it (Eqs. (5)–(6)), and its intensity (Eqs. (12)–(13));
4. draws the gaps and shortens them if they overflow (Eqs. (7)–(8));
5. places the storms (Eqs. (9)–(10)), draws each peak position, and computes the precipitation and aggregates.

5.3 Reproducibility

Two `numpy.random.RandomState` streams are seeded from the one seed: one draws the categories and base intensities, the other everything in a sequence. A fixed seed reproduces the batch exactly, apart from the identifiers and timestamps. Each sequence also carries a per-event seed for the storm–typhoon coupling, from a separate stream that does not disturb the storm draws.

5.4 Validation

`validate_event_set` checks every sequence:

- the type is one of the four, and the storm count is 1–5 and equals the storms held;
- there are $n - 1$ gaps, each at least its type's minimum;
- each storm starts after the previous one ends;
- the last storm ends by W_P, leaving at least W_{drain};
- every intensity category is valid;
- the total duration matches the durations and gaps.

Times are compared to within 10^{-9} h. Eqs. (6) and (8) make the timing checks hold by construction; validation is a second line of defence.

5.5 Command Line and Outputs

```
python phys.py port --stressm [--num-sims N] [--gauge-id GID] [--seed N]
```

`--num-sims` sets the count (default 10,000). The stage deletes the catchment's classifiers, which were trained on the previous catalogue, then runs the stress pipeline. That pipeline saves `storm_sequences.json` (the catalogue) and `sequences_summary.json` (batch statistics) for the catchment. The classifiers are then retrained per gauge from the Portfolio Stress screen.

6 Sensitivity Analysis

Every figure in this section is exact. It is computed in rational arithmetic from the tables above, with durations and gaps rounded as the code rounds them.

6.1 Sequence Types

For the default weights (Table 6), a sequence is isolated with probability $\sum_c w_c\,(1 - P_c(\text{sequence}))$. A multi-storm type has probability $\sum_c w_c\,P_c(\text{sequence})\,\pi_{c,t}$, where $\pi_{c,t}$ is its weight in Table 2.

Table 9: Expected sequence types in a default batch of 10,000.

Type	Expected count	Share
isolated	5,225	52.25%
doublet	1,469	14.69%
cluster	2,116	21.16%
persistent	1,190	11.90%

The doublet and cluster counts are 1,468.75 and 2,116.25 exactly. By storm count, a batch of 10,000 holds 5,225 sequences of one storm, 1,468.75 of two, 1,058.13 of three, 1,653.13 of four and 595 of five.

6.2 Duration Cap

The cap of Eq. (6) binds on **60.5%** of all duration draws in the default batch. It binds on every catastrophic duration in a five-storm sequence: the cap there is 26.4 h, and a catastrophic storm lasts at least 48 h. Table 10 gives each configuration.

Table 10: Share of duration draws capped (%), by configuration and category.

Type	n	Cap (h)	Moderate	Severe	Extreme	Catastrophic
isolated	1	156.0	0.0	0.0	0.0	30.3
doublet	2	66.0	0.0	25.2	76.1	97.5
cluster	3	36.0	38.4	91.0	100.0	100.0
cluster	4	21.0	89.6	100.0	100.0	100.0
persistent	4	34.5	45.2	93.6	100.0	100.0
persistent	5	26.4	75.7	99.6	100.0	100.0

6.3 Gap Shortening

Gaps are shortened (Eq. (8)) in **81.1%** of multi-storm sequences. That is 99.0% of moderate-to-severe persistent sequences and 100.0% of catastrophic ones (all but two in 10^{10}). By type, the rate is 39.5% of doublets, 99.6% of clusters and 99.5% of persistent sequences. In **63.4%** of multi-storm sequences every storm is capped, which holds every gap at its minimum (Table 11). A shortened gap is never shorter than its minimum.

Table 11: Multi-storm sequences whose gaps are shortened, and (in brackets) whose every storm is capped (%), by configuration and category. 100.0 is above 99.95.

Type	n	Moderate	Severe	Extreme	Catastrophic
doublet	2	0.9 (0.0)	47.9 (7.2)	92.9 (60.2)	99.7 (95.6)
cluster	3	96.4 (7.2)	100.0 (78.8)	100.0 (100.0)	100.0 (100.0)
cluster	4	100.0 (70.5)	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)
persistent	4	93.9 (4.2)	100.0 (76.8)	100.0 (100.0)	100.0 (100.0)
persistent	5	100.0 (24.8)	100.0 (98.2)	100.0 (100.0)	100.0 (100.0)

Most multi-storm sequences therefore take their gaps from the minimum of Table 4 rather than its full range. The gaps themselves are 6–36 h and 24–72 h only where the storms are short.

7 Validation and Backtesting

7.1 Structural Validation

`validate_event_set` finds no invalid sequence in a default batch, or in 400 sequences of every category with every forced type. Each sequence ends by hour 156, and every gap is at least its minimum.

7.2 Type Distribution

Pooled over 100 seeds of 10,000 sequences, every type's count is within 1% of Table 9. A single batch varies more: its persistent count has a standard error of about 3%.

7.3 Downstream Classifier

The classifiers' training summary on file (`training_summary.json`, 31 October 2026) covers 11 Thames gauges, with a mean AUC-ROC of 0.9935. Those classifiers were trained on v2.0's sequences, and so is the catalogue on file. Both predate v3.0, and its longer gaps change the hydrographs. The

next `--stressm` run regenerates the catalogue and deletes the classifiers, which must then be retrained. MKM-FC-001 owns that evidence. This document does not claim the classifiers' performance for v3.0's sequences.

7.4 An Example Gauge

GAUGE-1aaf8b08, in the same summary: AUC-ROC 0.9929, accuracy 0.9776, Brier score 0.0169; flood rate 8.6% over 1,713,600 training vectors. The dominant features are `log_h_s` (76%) and `log_t_end` (17%).

8 Model Limitations and Known Weaknesses

1. **The durations do not fit the window.** With Table 3's durations, three to five severe or worse storms and their minimum gaps cannot fit in 156 h. The cap binds on 60.5% of draws and truncates the most extreme multi-storm sequences, understating their duration. Recalibrating Tables 1 and 3 (fewer storms for long categories, or shorter storms in sequences) is outstanding.
2. **Gaps are mostly at their minimum.** In 63.4% of multi-storm sequences every gap is held at $g_{\min}$, so recovery between storms is the least the gap type allows rather than a draw from its range.
3. **Gap shortening is proportional, not physical.** The time above each gap's minimum is scaled uniformly.
4. **Storms are independent within a sequence.** Durations and peak positions carry no serial correlation, and a long first storm does not shorten the second.
5. **One footprint.** Every storm in a sequence has the same spatial footprint.
6. P_{base} **is fixed per catchment.** It does not vary by season or wetness, and a catchment without a storm module takes the Thames's value.
7. **The season is the generation date.** Each sequence's `sequence_start` is the time it was generated. The stress pipeline reads that month as the season for each gauge's base level, so every sequence in a batch uses the base levels of the month the batch was run.
8. **A control with no effect.** `FIRST_STORM_DOMINANT_PROB` does not change the output distribution (§3.7).
9. **Stationarity.** No climate scenario changes the weights, persistence or intensities.

9 Validation Questions

9.1 Q1 — Purpose

"Is the model being used for its intended purpose?"

Yes. It generates the stress catalogue that the stress pipeline turns into hydrographs for the classifiers and the Portfolio Stress scenarios (§2.2).

9.2 Q2 — Conceptual Soundness

"Is the model conceptually sound?"

The structure is sound: typed sequences in a fixed window, a guaranteed gap between storms, and correlated intensities. Its parameters are not mutually consistent, however, because the storms are too long for the window (§6.2, §8 item 1).

9.3 Q3 — Data Quality

"Is the data quality adequate?"

The parameters are expert-set, not fitted to observed sequences. No observed sequence catalogue has been compared with them.

9.4 Q4 — Developmental Evidence

"Is there adequate developmental evidence?"

Unit tests cover the samplers, the cap and gap minimum, the validator, the override path and reproducibility.

9.5 Q5 — Outcomes Analysis

"Are outcomes reasonable?"

The sequences are valid by construction and the type mix is as stated. Whether sequences like these occur over the Thames is untested.

9.6 Q6 — Ongoing Monitoring

"What ongoing monitoring is required?"

On each regenerated catalogue, monitor:

- the type counts against Table 9;
- the cap rate: a rate above 50% calls for recalibration of the durations, and the baseline rate is 60.5%;
- the gap-shortening rate by type: a rate above 70% says the gaps and durations are inconsistent with the window, and the baseline is 99.6% for clusters and 99.5% for persistent sequences;
- each retrained classifier's AUC-ROC, flagging a gauge below 0.95.

The cap and shortening thresholds are breached at the baseline, as §8 records.

9.7 Q7 — Benchmarking

"Has the model been benchmarked?"

Not against observed sequences or another generator. The implementation is benchmarked against Wolfram (Q8).

9.8 Q8 — Implementation

"Is the implementation verified?"

Wolfram oracle benchmark v0.2 tests every equation and claim in this document. It replays 19,600 sequences number for number, and checks the exact probabilities in Wolfram.

9.9 Q9 — Overall Risk

"Is the overall model risk acceptable?"

Acceptable for its Tier 2 use, with two outstanding items: the durations must be recalibrated so that multi-storm sequences are not mostly capped (§8 item 1), and the classifiers must be retrained on v3.0's catalogue (§7.3). Until then, the stress scenarios understate the duration of the most extreme sequences.

10 Document History

Date	Version	Description	Author
11-Mar-2026	2.0	First issue: the multi-storm sequence generator	David K Kelly
08-Nov-2026	3.0	Gap minima reserved before the duration cap, so every multi-storm sequence keeps its gaps; tables moved to config so an override reaches the generator; figures, correlation, base precipitation, paths and evidence stated as computed (Wolfram benchmark v0.1)	David K Kelly